Intel 与 NVIDIA 年报文本分析

AD698 Assignment 2 个人课程项目

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在第一次年报分析的基础上，我想进一步确认：引入词向量后，能否更好地区分Intel与NVIDIA的文本？除了分类，模型找到的关联词能否帮助查阅年报？为此，我比较了TF-IDF、Word2Vec和GloVe，并整理了它们的结果与局限。

完成内容

使用Python核对两家公司2020至2024年的30条章节记录和2,719个文本片段，检查字段缺失、片段编号和文本长度。分类比较使用1,624个片段，其中1,296个用于训练、328个用于测试。我还分析了liquidity、growth和risk的邻近词，用UMAP展示文本分布，并完成英文报告。

主要发现

方法	2024年样本准确率	识别正确的片段
TF-IDF 线性分类器	97.87%	321 / 328
Word2Vec + MLP	65.24%	214 / 328
GloVe + MLP	57.93%	190 / 328

这次运行中，TF-IDF的分类准确率最高，加入词向量并没有改善结果。因此，我建议保留TF-IDF用于公司识别，把词向量作为扩展检索词的参考。关联词仍需回到原文核查，不能直接当作财务结论。

比较边界：Word2Vec训练使用了未标注的2024年文本；不同流程对公司名称的处理及分类器也不相同，因此这些结果不能视为严格同条件下的方法优劣比较。

与申请岗位的联系

这项课程研究中，我需要按公司、年份和章节整理资料，核对样本与结果，再把分析写成图表和英文报告。我希望将这些经验用于业务支持中的数据核对、报表整理和资料维护。

附录

后附英文研究报告全文，保留原有的4幅图、4张表及参考文献。

Word Embeddings and the Geometry of Rivalry in Intel and NVIDIA Filings

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AD 698 Applied Generative AI

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October 14, 2026

Business Question and Evaluation Design

Based on this experiment, TF-IDF remains the most powerful company classifier. I compared Intel and NVIDIA in the semiconductor industry (NAICS 3344) to test whether semantic word vectors could improve discriminative power and help analysts interpret their annual reports. The reused corpus contains 30 section records and 2,719 text chunks from filings submitted between 2020 and 2024. The classification task uses the same 1,624 sampled word blocks as in Assignment 1. Of these, 1,296 from 2020–2023 were used for training, and 328 from 2024 were used for testing.

The in-house Word2Vec model employs a skip-gram architecture with 100 dimensions, a 5-word window, a minimum word frequency of 3, and is trained for 10 epochs (Mikolov et al., 2013). GloVe uses pre-trained 6B, 100-dimensional vectors (Pennington et al., 2014). The mean-pooled word blocks were fed into a two-layer multilayer perceptron (MLP) with 64 hidden units, a ReLU activation function, and a dropout rate of 0.2. Both models were trained for 30 epochs using the AdamW optimizer with a learning rate of 0.001, and the cross-entropy loss function was employed.

Q1 Model Comparison

Table 1

Accuracy on the Same 328 Test Chunks

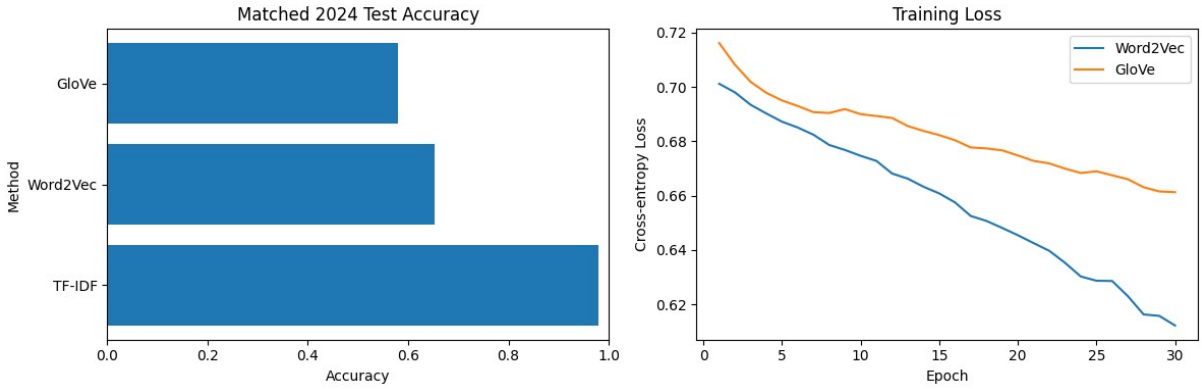
Method	Accuracy	Correct
TF-IDF BoW linear classifier	0.9786585366	321
Scratch Word2Vec MLP	0.6524390244	214
Pretrained GloVe MLP	0.5792682927	190

Word2Vec falls 32.621951 percentage points below TF-IDF; GloVe falls 39.939024 points below it. GloVe is 7.317073 points below scratch Word2Vec. Moving to these embedding pipelines therefore did not improve company discrimination.

Q1 Interpretation and Reliability

Figure 1

Company Classification Accuracy and MLP Training Loss



The loss curve at the 30th epoch continues to decline, particularly for Word2Vec. When combined with simple mean pooling and different classifier architectures, this makes it impossible to attribute the accuracy gap entirely to the quality of the word vectors. The General English version of GloVe did not provide a classification advantage here, but this does not prove that the languages in the report do not fall within the scope of General English.

Following the assignment example, Word2Vec was trained on the entire corpus, including unlabeled text from filing year 2024. This makes the comparison transductive rather than a strictly prospective evaluation. Assignment 1 removed company identifiers, whereas the embedding preprocessing retained them. These differences mean that the accuracy comparison does not isolate the effect of the text representations alone.

Nearest Neighbor Evidence

Figure 2

Selected Word2Vec Queries for Liquidity and Growth

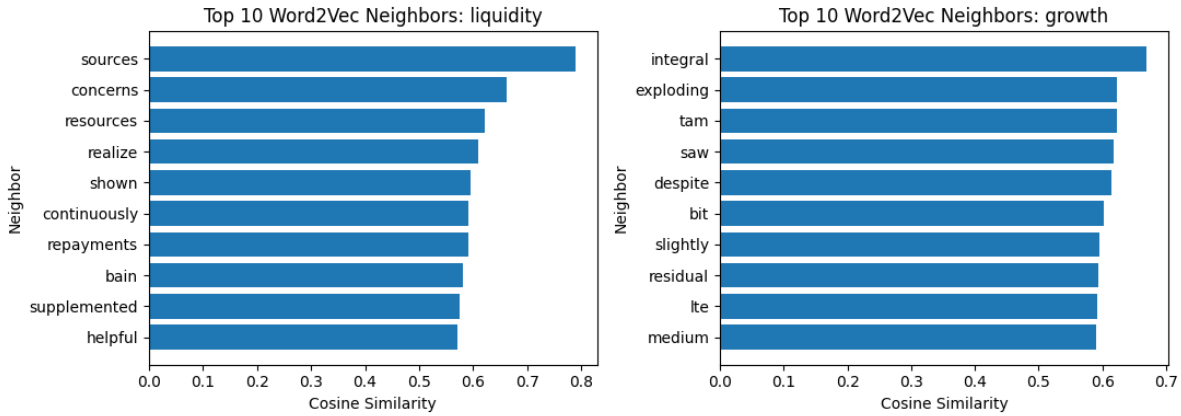


Table 2

Separate Company Models and the Top Five Neighbors of Risk

Rank	Intel	Similarity	NVIDIA	Similarity
1	protect	0.694578	exposed	0.756099
2	fluctuations	0.692804	statement	0.708103
3	mitigate	0.690397	free	0.689452
4	degree	0.686150	concentrations	0.688772
5	concentrations	0.668077	attributed	0.671812

Liquidity is closest to sources (0.789833), while growth is closest to integral (0.669669). Other neighbors, like continuously and despite, show why contextual proximity should not be read as a synonym list. Intel risk neighbors include protect, fluctuations, and mitigate; NVIDIA's include exposed, statement, and free. These suggest search terms to investigate; they are not verified differences in financial exposure. The two company models have separate coordinate systems; therefore, I compare their neighbor lists rather than subtracting their vectors.

Q2 Embedding Geometry

Figure 3

GloVe UMAP Projections by Company and Section

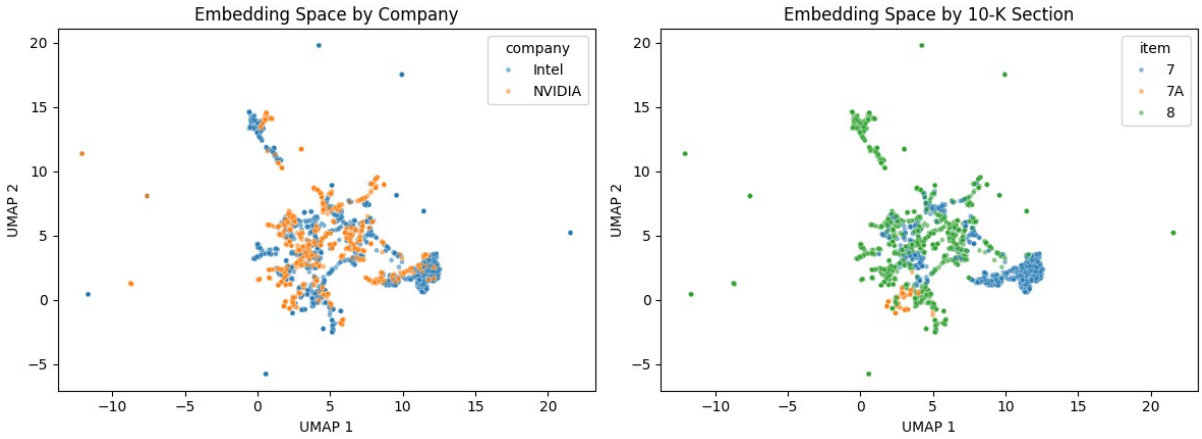


Table 3

Company Separation Within Each Section

Item	Cosine silhouette in 100D	A1 section accuracy
7	0.110324	1.000000
7A	0.075289	1.000000
8	−0.006096	0.962500

As shown in Figure 3, there is significant overlap between the company colors, while the regional colors reveal that Item 7 exhibits a more distinct branch in the lower-right direction; Item 8 clearly occupies most of the central and upper-left regions of the structure, whereas Item 7A appears as a small cluster near the center.

In the original GloVe space, Item 7 had the highest company separation score (cosine silhouette score 0.110324), followed by Item 7A (0.075289), while Item 8 exhibited a very high degree of overlap (−0.006096). This broadly aligns with A1: in A1, Categories 7 and 7A tied for first place with an accuracy of 1.000000, while Category 8 scored 0.962500; however, UMAP is

an unsupervised projection and cannot reproduce supervised classification boundaries (McInnes et al., 2018).

Note. The silhouettes measure company separation within each section, not separation between section labels. The executed UMAP emitted spectral initialization warnings and fell back to random initialization; no runtime exception was recorded. The chart is exploratory.

Annual Returns and Market Context

Table 4

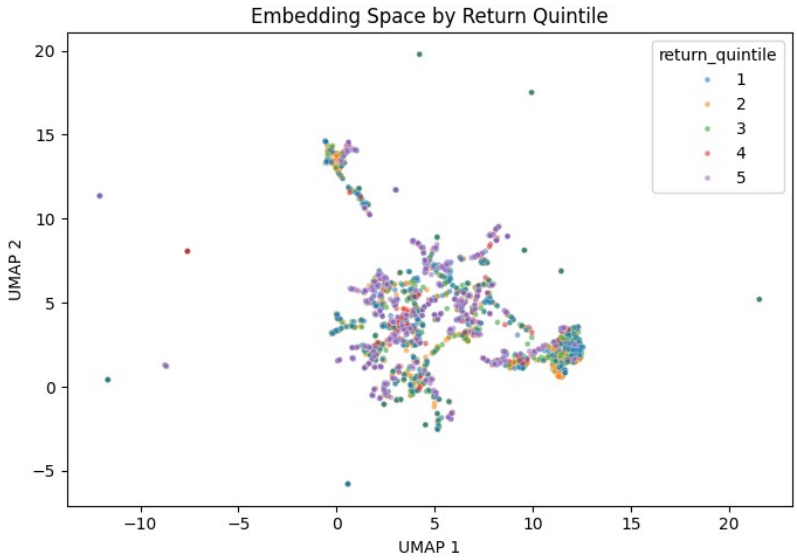
Annual Adjusted Stock Returns and Pooled Return Quintiles

Year	Intel return	NVIDIA return	Intel quintile	NVIDIA quintile
2020	-0.146888	1.222956	2	4
2021	0.060525	1.254821	3	4
2022	-0.466427	-0.502646	2	1
2023	0.945624	2.390172	3	5
2024	-0.595651	1.712492	1	5

Note. Returns are expressed as decimals: 1.000000 equals 100%. Quintiles are based on the ten unique company-year returns, with 1 representing the lowest returns and 5 the highest. Historical stock prices were obtained from Yahoo Finance using the yfinance Python package (Aroussi, n.d.). Annual returns use adjusted closing prices from the last trading day of each year and the previous year. Calendar-year returns are matched to SEC filing years, not fiscal reporting periods.

Figure 4

Embedding Space by Return Quintile



The return-quintile colors overlap substantially, and the highest-return group does not form an isolated region. Although the plot contains 2,719 text chunks, the return labels represent only ten unique company-year observations from two companies. Repeating these labels across chunks does not increase the number of independent return observations.

Q3 Analyst Efficiency and Business Recommendations

As an analyst, one could use the nearest-neighbor algorithm to expand the search for funding sources, growth, or market risk and compare the contextual vocabulary of two companies. Then, considering the relevant results for liquidity, growth, and risk—which provide starting words for locating specific paragraphs—one must also keep in mind that words with similar learned contexts such as “continuously” and “despite” require manual screening.

Analysts may overlook negations, conditions, monetary amounts, time references, and changes in management estimates. Nearest-neighbor terms also cannot distinguish between actual losses and hypothetical risk exposures. It is still necessary to read the source text paragraphs before drawing financial conclusions.

An average text block contains 764.659801 characters. Assuming 3,000 characters per page, one average text block is equivalent to 0.254887 pages; ten average text blocks are equivalent to 2.548866 pages. Therefore, if the search results lead an analyst to review ten relevant paragraphs, their attention can be focused on approximately 2.55 pages of content. However, since this implementation returns ten words rather than ten text segments, this is only a conditional estimate based on reading volume, not the actual number of pages replaced. Furthermore, overlapping text blocks will result in duplication of source text content.

Figure 4 shows severe color mixing between high and low return rates, and the sorting of return rates is unclear; the quintiles in View 7A are also intermingled. For this application, I would opt to use embeddings for search term expansion while retaining TF-IDF for company differentiation. Before applying the semantic results to investment monitoring, I would first validate the source text paragraphs, standardize the preprocessing workflow, train the embeddings using only data from earlier years, and test the system with a broader range of companies and years. Based on the available evidence, this approach is supported as a navigation aid for analysts.

References

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McInnes, L., Healy, J., & Melville, J. (2018). *UMAP: Uniform manifold approximation and projection for dimension reduction.* arXiv. https://arxiv.org/abs/1802.03426

Mikolov, T., Chen, K., Corrado, G.S., & Dean, J. (2013). *Efficient estimation of word representations in vector space.* arXiv. https://arxiv.org/abs/1301.3781

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